Started on	Wednesday, 23 October 2024, 1:45 PM
State	Finished
Completed on	Wednesday, 23 October 2024, 2:15 PM
Time taken	29 mins 49 secs
Marks	2.30/3.00
Grade	**2.30** out of 3.00 (**76.67**%)

Information

Information

This page contains all the problems for this test. The very last problem asks you to contact the person in charge of the exam and tell him or her the 4-digit key given in the problem text. In return you will be given a 5-digit signing code which you must give as the answer to the problem.

This problem does not count towards the final score, but **tests missing this code will not count towards the final grade**.

The following rules apply:

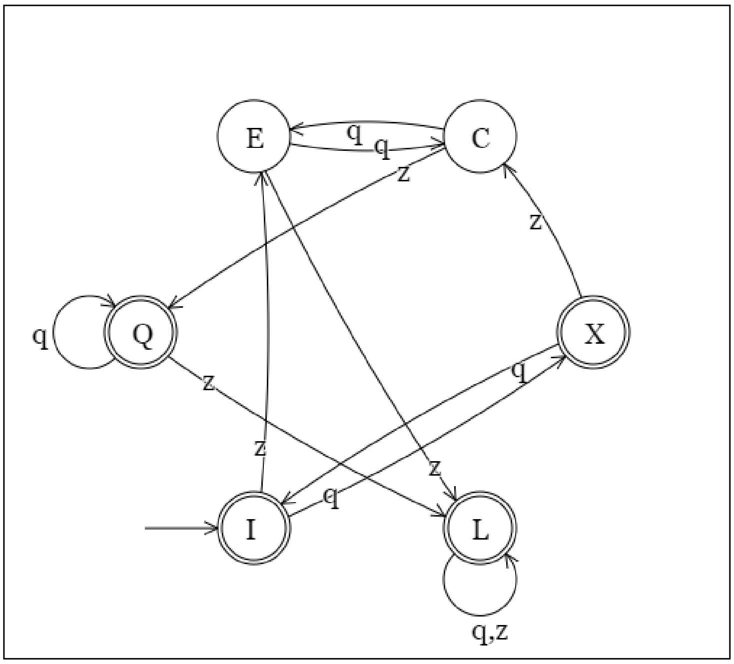
- Total time allowed: 30 minutes. The test will automatically close if time runs out.
- UiA's usual rules in regards to cheating on exams apply.

Question **1**

Correct

Mark 1.00 out of 1.00

Minimize the following DFA.



Write the answer as a set of sets of equivalent states.

{{"Q", "L"}, {"I", "X"}, {"E", "C"}}

Your last answer was interpreted as follows:

$$\{\{Q, L\}, \{I, X\}, \{E, C\}\}$$

Question **2**

Partially correct

Mark 0.30 out of 1.00

Which temporal logic statement matches the English sentence "It is always possible that darkness leads to a later light."?

G(darkness ⇒ F light)

Question **3**

Correct

Mark 1.00 out of 1.00

Given the set $[1, 2, 3, 4, 5]$, and relations $\{[2, 5]$, $[2, 2]$, $[3, 1]$, $[1, 3]\}$.

1. Fill in the relations matrix below such that all given relations are marked by a "**1**" in the correct position.
2. Then extend the set of relations by adding mandatory "**1**"'s so that the final set of relations is an equivalence relation.
3. All other entries in the matrix must be marked "**0**".

-	1	2	3	4	5
1	1	0	1	0	0
2	0	1	0	0	1
3	1	0	1	0	0
4	0	0	0	1	0
5	0	1	0	0	1

Your last answer was interpreted as follows:

$$\begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

Question **4**

Correct

Mark 0.00 out of 0.00

Signing code

Before closing the test you must answer this problem with a signing code given to you by the person in charge of the test.

Tests missing this signing code will be ignored and will not count towards the final score.

Key: 512

Signing code: 33568

Your last answer was interpreted as follows:

$$33568$$

◄ Test 1 (topics 1-3: Introduction, Concepts, Induction, Recursion, Grammars)

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